

1 General

1.1 Approach to Layout

1.1.1 Provide robust power and GND planes

1.1.2 Labeling:

1.1.2.1 Connector/Headers clearly labeled

1.1.2.2 Pin 1 clearly identified

1.1.2.3 Power and GND pins labeled

1.2 Mis-wiring Tolerance

1.2.1 Circuitry to prevent damage for reversed polarity on power connection

1.3 Software Development Language Supported

1.3.1 MicroPython

1.3.2 CircuitPython

1.3.3 C/C++

2 Functions/Capability

2.1 Power

2.1.1 Board Main Power: 5.5 to 24V DC @ max 10A RMS

2.1.2 OTS 5V DC to DC converter 5.5 to 24V input, 3A output

3 3.3V 1.5A linear regulator

3.1 Controller: Pico 2W

3.1.1 Dual core

3.1.2 WiFi/BLE

3.1.3 C/C++, MicroPython, CircuitPython support

3.1.4 LED pin level indicators

3.1.5 0.1" receptacle (2x20) module plugin

3.1.6 0.1" pin header (2x20) for pin access

3.2 Visual Indicators

3.2.1 5V

3.2.2 3V3

3.2.3 GPIO LED state indicator LED (28)

3.3 Motor Control Capabilities

3.3.1 2 x Pololu VNH5019 Motor Driver Carrier as a daughter Cards supported

3.3.2 2 x onboard brushed Motor circuitry (not populated)

3.3.3 2x stepper motor support circuitry (not populated)

3.3.4 2 x Servo motor Interfaces (3 pin 0.1" header)

3.3.4.1 PWM

3.3.4.2 GND

3.3.4.3 Jumper selectable for 3V3 or 5V power

3.4 Sensor/Accessories Interfaces

3.4.1 Ultrasonic sensor interface (4 pin 0.1" header)

3.4.1.1 Trigger

3.4.1.2 ECHO

3.4.1.3 GND

3.4.1.4 Selectable 3V3 or 5V

3.4.2 Qwicc (Sparkfun)/STEMMA QT (Adafruit) Connector (4 pin)

3.4.2.1 GND

3.4.2.2 3V3

3.4.2.3 I2C data

3.4.2.4 I2C clk

3.5 ADC capabilities

3.5.1 3 general ADC inputs

3.5.2 Analog divider and filter component for each interface

3.5.3 3 pin 0.1" header per ADC Interface

3.5.3.1 Analog input

3.5.3.2 GND

3.5.3.3 jumper selectable 3V3 or 5V

3.5.4 Jumper selectable to on board current sensing or external circuitry

3.6 Mechanical

3.6.1 Type down for USB-uB extension cable from Pico 2W

3.6.2 Minimum of insulated mounting holes for 3mm fasteners

3.6.3 Keep out areas for VNH5019 Motor Driver Carrier Supports